Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Outline

Part 1

Setting and reforms

Question and headline

Setting. Sequential electricity wholesale markets. Spain switched the trading time resolution from 60 minutes to 15 minutes in two staggered reforms in 2025, March and October.

Question. Does finer time-resolution reshape strategic bidding? Does market power go up or down? What effect do we see in prices, market differentials, and imbalance penalties?

Answer. COMPLETE!!!

Caveat. An April 2025 blackout triggered a separate regulatory regime that inflated the system-cost line. Granularity and that regime are calendar-overlapping but mechanistically distinct.

The institutional setting — sequential electricity wholesale markets

The product. Delivery of MW (rate) over a fixed short time slot, by different technologies.

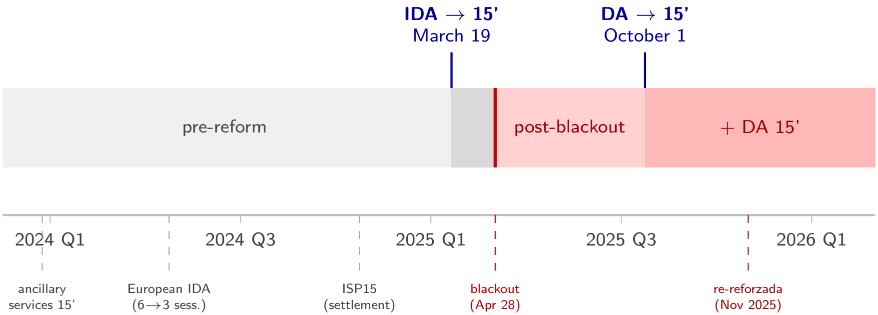
Sequence of markets. Electricity cannot be stored cheaply, supply must equal demand in real time. For this reason, we have sequential **wholesale** markets before delivery:

- **Forward auction ("day-ahead", DA)** — clears the day before. Biggest market, with the most liquidity.
- **Spot auctions ("intraday auctions", IDA)** — clears closer to delivery. Corrections.
- **Balancing (real time)** — corrections by the system operator (REE). Highly complex, with separate pricing mechanisms (APPENDIX).

Both DA and IDA are **uniform-price multi-unit auctions**; bidders submit step-function supply and demand curves and price is set at the intersection of the aggregate supply and demand curves, using a complex algorithm that maximizes social welfare.

Huge complexity: ancillary markets, bilateral contracts, cross-border trading, continuous intraday markets, etc. We focus on DA and IDA, and present basic descriptive statistics on some of the rest.

The Spanish reform sequence



- **Main reforms.** Change in time granularity of **IDA** (March 2025) and **DA** (October 2025): from 60-min products to 15-min.
- **Other pre-reforms in 2024**: ancillary services, European IDAs, imbalance quarter-hourly settlement.
- **April 2025 blackout** that triggered separate regulatory regime ("operación reforzada", later "re-reforzada").

Literature review

Sequential markets, forward contracts, Cournot. Allaz and Vila (1993); Ito and Reguant (2016).

Empirical bidding in multi-unit electricity auctions. Hortaçsu and Puller (2008); Hortaçsu et al. (2019).

Trading-granularity and settlement-period reforms. Märkle-Hußet al. (2020); Csereklyei and Khezr (2024).

Renewable price impacts and ancillary services in Spain. Brown and Reguant (2026).

Iberian blackout and post-blackout regime. Daví-Arderius and Graf (2026); ENTSO-E Expert Panel (2026); CNMC (2026).

Part 2

Mechanisms and identification strategy

A priori mechanisms expected from the reform sequence

- We expect production to shift to the granular side of the market.... MORE FOR RENEWABLES. This effect would be bigger for ID15 into the IDAs, since it is near delivery, and greater the bigger the shocks volatility between DA and IDA are.
- If residual demand of the price setter firms gets more inelastic in each quarter (thinner market), we expect prices to increase. If it gets more elastic, the opposite
- CHANGE IN THE BID SHAPE WITH THE REFORM: MORE VARIATION IN TRANCHES, MORE DETAIL, LESS AGGREGATION.

Data architecture

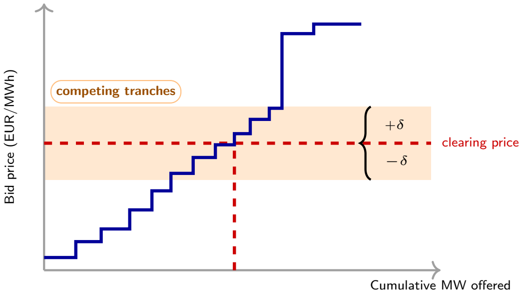
I build from scratch a **full automated data pipeline** for the public market data, using 3 sources:

- Wholesale market operator data (OMIE): DA and IDA prices, auction quantities, bilaterals, accepted bid-curves, etc. Continuos market is included too, but not the focus.
- System operator data (REE) via their API: balancing market prices, ancillary services, etc.
- European data portal (ENTSO-E) via API: cross-border flows, wind and solar production, etc.

Magnitude. 420GB of raw data, 200GB processed in *parquet* format. Around 10 billion rows. Dedicated external SSD for storage and processing.

Time window. January 2018 – today, idempotent code. Different granularities across time.

Bid-curve variables



Not all tranches of the bid curves are equally relevant for strategic bidding, we focus on **tranches around clearing price in that period**, using a bandwidth:

$$\mathcal{B}_{u,d,p} = \{k : |p_k - \mathrm{MCP}_{d,p}| \leq \delta\}.$$

choosing the bandwidth $\delta = p_{90} - p_{50}$ in the time-window of interest. Note that both the level and the width of $\mathcal{B}_{u,d,p}$ vary across time!

Bid-curve shape measures:

- Slope and level of the bid curve: for each curve, fit $p_k = \hat{\alpha} + \hat{\beta} Q_k$ in $k \in \mathcal{B}$ and get $(\hat{\alpha}, \hat{\beta})$. Similar with the 2nd order polynomial to get the curvature $\tilde{\gamma}$.
- Robust measures to few tranches inside the band:
  - Quantity-weighted Herfindahl over tranches: $\mathrm{HHI}_p = \sum_{k \in \mathcal{B}} w_k^2$, with weights $w_k = \frac{q_k}{\sum_{j \in \mathcal{B}} q_j}$.
  - Quantity-weighted SD of tranche prices: $\sigma_p = \sqrt{\sum_{k \in \mathcal{B}} w_k (p_k - \bar{p})^2}$, where $\bar{p} = \sum_{k \in \mathcal{B}} w_k p_k$.

Residual demand and hour classification

Key idea. Granularity is useful when there is variation in the residual demand for each firm, and particularly for the strategic bidders. Renewable producers exploit granularity due to operational reasons (forecast updates).

We calculate the residual demand as the thermal gap: final program net of realized wind and solar generation, $\mathsf{RD}(q,h,d) = [\text{load} - \text{wind} - \text{solar}]_{q,h,d}$, observed at quarter-hour resolution since 2024. To make the comparison scale-free we use the **coefficient of variation** $\mathrm{CV}(h) = \mathbb{E}_d[\mathsf{sd}_q(\mathsf{RD}(q,h,d))]/\mathbb{E}_d[\overline{\mathsf{RD}}(h,d)]$, and split the hours of the day into three classes:

- $\left.\begin{array}{l}\textbf{Morning ramp } \mathcal{C}_M\text{: } \mathrm{CV} \approx 6.0\% \ (4.0 \times \text{ flat}) \\ \textbf{Evening ramp } \mathcal{C}_E\text{: } \mathrm{CV} \approx 3.8\% \ (2.5 \times \text{ flat})\end{array}\right\}$ **Critical** $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$, $\overline{\mathrm{CV}} \approx 4.6\%$ $(3.1 \times$ flat)
- **Flat** $\mathcal{F}$, deep night hours, $\mathrm{CV} \approx 1.5\%$ (baseline).
- **Midday** $\mathcal{M}$, $\mathrm{CV} \approx 5.7\%$ $(3.8\times$ flat).

Identification strategy: bid-curve shape

Identification for bid-curve shape. Check evolution of bid-curve shape measures in critical hours vs flat hours, pre vs post reform, to measure the effect of granularity on bid shape.

Why use a DiD here? It differences out across-hour changes in bidding complexity, such as more advanced software, prediction models, etc.

Caveat: flat hours are not a perfect controls. They are also affected by the reform, but economic theory tells us that the biggest effects of the reform are in critical hours → **lower bound of the effect.**

The identification assumption is the usual **parallel trends:** absence granularity reform, the evolution of the bid shape of critical and flat hours would have been the same.

Note that our bid-curve shape measures are robust to seasonal level changes! Main threat: post-reform shocks that affect differently critical vs flat hours, not present before.

Identification strategy: level variables

For level variables like prices, quantities, etc, **seasonality** is the big issue, since changes in renewable production affect differently critical and flat hours across the year.

Moreover, for quantity variable we have to be very very catious and interpret correctly the different programs and REE restrictions.

Following previous work of the EPEX reform (Märkle-Hußet al., 2020) we use Bayesian Structural Time Series (Brodersen et al., 2015) and standard OLS exploiting all the rich data and granularity.

The most important part is the choice of windows and controls:

- **Windows.** We use a long pre-window from 2022 to capture full seasonality across years. The other alternative is to use reforzada-constant windows, or excluding periods with different regimes but that would reduce the pre-window length and thus the ability to capture seasonality and real renewable growth.
- **Controls.** We control for wind and solar production, and gas price, which are the main drivers of price variation in the Spanish market. We also include year-by-year interactions to capture the growth in renewables, which crucial to get depressing effect in prices of solar in midday hours.

Part 3

Empirical results

Summary of our results

For ID15:

- Decrease in prices in both IDA and DA after controlling for renewable penetration and seasonality. Price wedge between DA and IDAs increase.
- No change in aggregate volume, wind sells more actively in IDA. Nuclear was out of the market post-ID15 and pre-blackout. SELL SHARE HERE.
- BID SHAPE HERE. RESIDUAL DEMAND HERE.

For DA15:

- Small decrease in DA prices, null in IDA, after controlling for renewable penetration and seasonality. Price wedge do not vary.
- AGGREGATE QUANTITIES HERE. TECH VARIATION HERE. SELL SHARE HERE.
- BID SHAPE HERE. RESIDUAL DEMAND HERE.

Prices — robust drop at ID15, null at DA15

	ID15 (IDA reform)		DA15 (DA reform)	
	IDA (own)	DA (cross)	DA (own)	IDA (cross)
OLS daily — base (month + DOW)	-32^{***}	-33^{***}	-8	-8
OLS daily — + year-by-renewable (per year)	-46^{***}	-47^{***}	$-\mathbf{9}$	$-\mathbf{10}$
OLS daily — + year-by-renewable (2024+25 pooled)	$-\mathbf{39}^{***}$	$-\mathbf{39}^{***}$	-5	-6
OLS hourly — base (hour + month + DOW)	-34^{***}	-34^{***}	-10^{*}	-10^{*}
OLS hourly — + year-by-renewable (per year)	-40^{***}	-39^{***}	-3	-4
OLS hourly — + year-by-renewable (2024+25 pooled)	$-\mathbf{30}^{***}$	$-\mathbf{29}^{***}$	3	2
BSTS daily	-34^{*}	-34^{*}	8	1

Window 2022-01-01 onward: preblackout for ID15 (40 days), 90 days for DA15. **Base controls** *(all rows): wind, solar, gas, calendar-month FE, day-of-week FE. Newey-West HAC SE. We average quarter-hour prices.*

- **ID15 own-venue price drop is robust** at −30 to −46 EUR/MWh across all specs; cross-market response on DA is the same magnitude (anticipation through backward induction).
- **DA15 own-venue null** (−9 to +8); cross-market also null.
- **ID15 effect persists at the 90-day post-window, including blackout** (OLS pooled -42^{***}, BSTS real effect −35 EUR/MWh) — full sensitivity at 40d / 90d / 180d / 196d in appendix.
- **Effect is broadly uniform across hour classes at ID15**: evidence of hydro-pump and batteries arbitrage (CHECK BID SHAPE!!!).

Price wedges — per-IDA-session levels and wedge SD

(a) Wedge level per session (EUR/MWh; $p_{DA} - p_{IDA,s}$; pre 2024-06-14+, Newey-West HAC).

	ID15 (IDA reform)			DA15 (DA reform)		
	S1	S2	S3	S1	S2	S3
OLS daily — base (month + DOW)	2.8^{**}	-0.1	-9.6^{**}	0.3	-2.8	2.8
OLS daily — + year-by-renewable (per year)	2.9^{**}	-0.4	-7.7^{*}	-0.1	-3.6	2.3
OLS daily — + year-by-renewable (2024+25 pooled)	3.0^{**}	0.3	-7.0^{*}	0.0	-2.4	3.7
OLS hourly — base (hour + month + DOW)	2.7^{**}	0.7	-0.9	1.0	-3.0	0.6
OLS hourly — + year-by-renewable (per year)	2.8^{**}	1.0	-0.7	0.6	-3.1	1.4
OLS hourly — + year-by-renewable (2024+25 pooled)	2.8^{**}	1.0	-0.7	0.6	-3.1	1.4
BSTS daily	1.6^{**}	1.5	-0.6	0.3	0.3	0.0

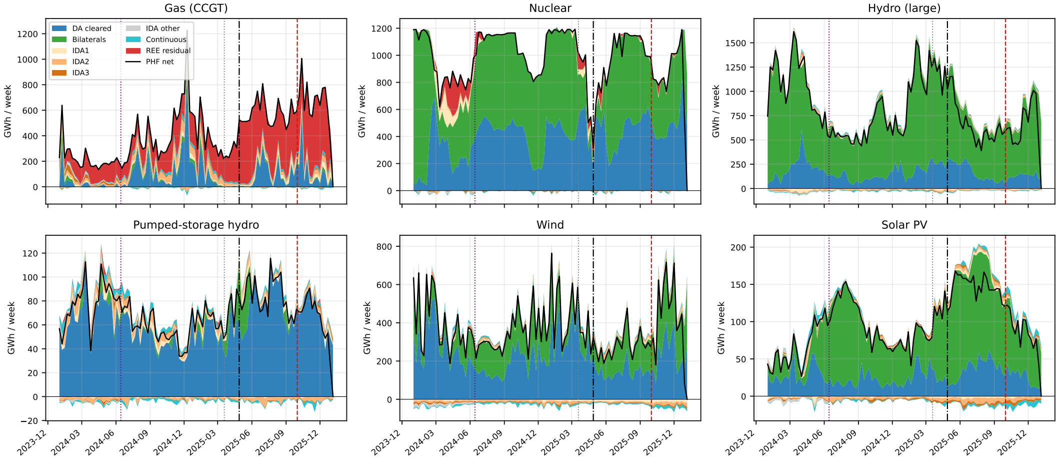
(b) Wedge SD effect, by hour class (BSTS, EUR/MWh).

	Overall	Morning ramp (5–8)	Midday (11–14)	Evening ramp (16–22)	Flat (1–3)
ID15	$\mathbf{1.1}^{**}$	$\mathbf{2.3}^{***}$	-1.6^{**}	1.2^{*}	-0.0
DA15	-0.1	0.5	-0.7	-1.4	-0.5

ID15: S1 wedge widens +2.7 to +3.0 (robust); S3 daily narrowing does *not* survive hourly or BSTS. Wedge SD jump concentrates in the **morning ramp** ($+2.3^{***}$) more than the evening ramp ($+1.2^{*}$); midday *compresses* (-1.6^{**}).
DA15: no per-session restructuring and no SD effect. INCLUDE PLACEBOS IN APPENDIX, THEY ARE ROBUST!!

Changes in quantity — programs must be read separately

Per-tech weekly program cascade (chronological, supply above 0, consumption / down-redispatch below 0). PDBC → Bilaterals → IDA1/2/3 → Continuous → REE post-IDA RT. Black line = PHF net.



Changes in quantity – main changes

WE SHOULD SAY QUANTITY CHANGES IN WORDS AND INCLUDE THE REGRESSIONS IN THE APPENDIX
MENTION SELL SHARE, DROP FIGURE, NOT NEEEDED.

Changes in bid shape – critical vs flat

HERE FOCUS ON CCGT, HYDRO PUMP AND CHP TOO (QUITE BIG)!! PUMP STORAGE AND HYBRID ARE SMALLER BUT INTERESTING, ALSO COAL.

Changes in bid shape

MORE HERE

Changes in residual demand slope for dominant firms

Object. For each of the four dominant generators, the slope $b_{m,t}$ of the residual demand it faces (Cournot primitive). Steeper $\Rightarrow$ less unilateral market power.

DiD spec (per firm, per reform × venue).

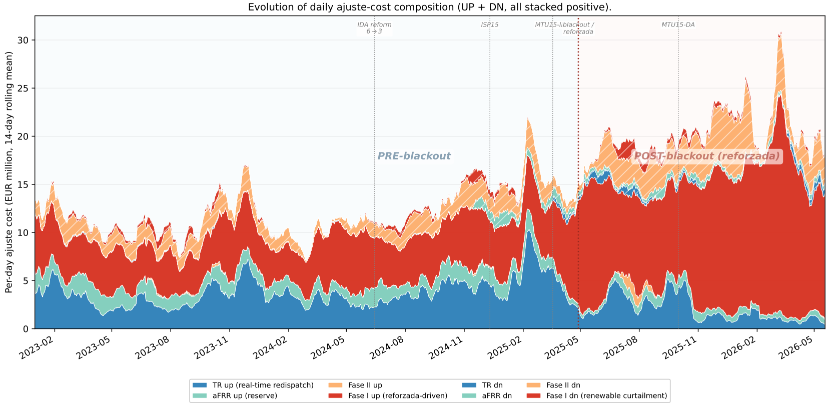
$$\log b_{f,t} \;=\; \theta_f \, \mathbb{1}\{t \in \text{post}\} \;+\; \text{month FE} \;+\; (\text{wind, solar daily}) \;+\; \varepsilon_{f,t}.$$

Table reports θ_f (% change in slope).

Change in slope per firm	Firm 1	Firm 2	Firm 3	Firm 4
March reform, slope on IDA	10%	13%	**17%***	13%
October reform, slope on IDA	11%	13%	12%	**15%*****
October reform, slope on DA	−2%	−1%	−2%	−3%

- **Main finding (green):** slope thickens on the IDA side at both reforms. Direction consistent for every firm; wind/solar daily-production controls barely move the coefficient $\Rightarrow$ not a supply-shift artefact.
- **Negative finding (red):** DA at October. The newly-reformed DA does NOT thicken its own slope. Effect is venue-specific to the IDA side.

Elephant in the room: ancillary services



- The system operator is paying CCGT to produce more through adjustments in the wholesale markets, and also to solar to... stop producing.
- This costs have increased substantially! QUANTITY Re-reforzada just after DA15 calls CCGT even before in the program cascade (TR up).

Wedge — mean unchanged, SD jumps, per-session restructures (ID15)

(a) BSTS: within-day wedge SD (Spec A, 2024 same-calendar placebo).

Wedge SD (EUR/MWh)	**Real**	**2024 plb.**
March, overall	**1.15****	-0.96
March, peak hours	**2.12*****	-0.68
October	1.12 (persists)	0.09

Wedge SD jumps at March, stays elevated through October — bounded per-product arbitrage: with 4$\times$ more delivery products, capacity per product binds. SD lands at $\sim 2\times$ the unconstrained benchmark, not zero.

Mean unchanged at either reform (granularity does not collapse the level wedge).

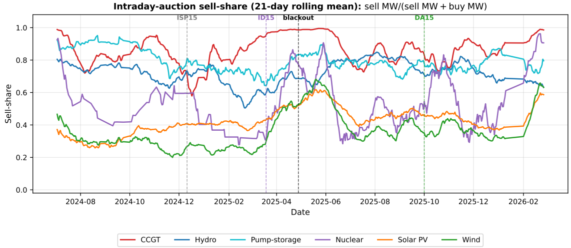
(b) OLS: wedge level per IDA session (Newey-West HAC, long pre 2024-06-14+).

Wedge	**ID15** $\hat{\theta}$ (EUR/MWh)	**DA15** $\hat{\theta}$
Aggregate	-0.04	1.31
Session 1 (early)	**3.0****	-0.09
Session 2 (mid)	0.3	-3.6
Session 3 (late)	$-$**7.7***	2.3

At ID15 the wedge restructures within day: S1 (early, nearer DA gate) widens $+3$; S3 (late, nearer delivery) narrows -7.7. Aggregate null.

No per-session restructuring at DA15.

Quantity — IDA sell-share rises at ID15 (figure + BSTS)



	CCGT	Hydro	Pump-storage	Nuclear	Solar PV	Wind
IDA \% dual sessions	30.6\%	43.3\%	36.1\%	19.7\%	58.8\%	82.7\%

IDA sell-share by technology, 21-day rolling. Vertical: reform cutovers. Generators submit BOTH buy and sell orders within the same IDA session ⇒ they actively arbitrage their own DA position. The level shifts at March 2025.

BSTS sell-share change at ID15 (pp) — long-pre Spec A with renewables and gas controls.

Technology	$\hat{\Delta}$ (pp)
Nuclear	**33.5*****
Wind	**23.5*****
Hydro	4.8***
Solar PV	3.6**
Pumped hydro	3.9
CCGT	−3.1

Nuclear and wind swing into selling. Pre-reform they used IDA as a one-way buy market; post-March they actively price output at 15-min resolution. CCGT (strategic-bidder benchmark) does NOT shift its DA/IDA mix.

Cleared quantities — OLS per technology, both reforms

OLS spec. Daily cleared GWh per (tech, market). Same RHS as the price spec; Newey-West HAC. Two cells per tech: *ID15 IDA* (granular venue at the IDA reform) and *DA15 DA* (granular venue at the DA reform).

	ID15 IDA cleared (PIBCI)		DA15 DA cleared (PDBC)	
Tech	$\hat{\theta}$ **(GWh/day)**	**p**	$\hat{\theta}$ **(GWh/day)**	**p**
CCGT	$-$**5.5*****	0.002	14	0.24
Hydro	$-$**3.8****	0.02	$-$**8.5*****	0.000
Pumped hydro	-0.6^{*}	0.05	**3.4****	0.003
Nuclear	3.7	0.62	**27***	0.07
Solar	**6.3*****	0.000	**6.5*****	0.000
Wind	**3.9*****	0.008	**11****	0.015

- **Renewables flow toward the granular venue at each reform.** ID15: solar +6.3 and wind +3.9 GWh/day INTO IDA. DA15: solar +6.5, wind +11 GWh/day INTO DA. Direction matches the model's aggregator-on-granular-leg prediction.
- **Dispatchable techs lose share on the IDA at ID15:** CCGT -5.5 and hydro -3.8 GWh/day. At DA15 the DA side scales up across the board (pump-storage, nuclear, wind).
- BSTS cross-check: same qualitative pattern (CCGT IDA -8, solar IDA +6.7, wind IDA +5.3 at ID15; solar DA +11 at ID15; CCGT DA +36, wind DA +18 at DA15).

Penalty — imbalance per MWh drops; aggregate volume does not

Setup. Parties that deviate from contract pay a regulated penalty per MWh of deviation.

BSTS spec. $\hat{\Delta} = \frac{1}{T_{\text{post}}} \sum_{t \in \text{post}} \left(Y_t - \hat{Y}_t^{\text{cf}} \right)$ on daily outcomes, with (wind, solar, gas) regressors and 2022−present seasonality.

Outcome	March reform	October reform
Penalty per MWh of deviation	$\mathbf{-33\%}^{***}$	flat
Aggregate imbalance volume (MWh/day)	-0.1%	-4%

- **Main finding (green):** penalty per MWh falls by a third at March. The granular spot lets the operator price-discriminate the residual imbalance at a lower marginal cost.
- **Negative findings (red):**
  - Aggregate imbalance volume does NOT drop $\Rightarrow$ granularity reorganises the *within-hour* composition of imbalance, not the size of aggregate forecast errors.
  - Penalty per MWh is flat at October $\Rightarrow$ the gain was captured at the IDA reform; the DA reform adds no further compression.

Bid shape — CCGT ladders widen in critical hours (Spec C)

DiD spec. For each (reform, venue) cell, per-curve fit:

$$Y_{u,d,p} = \theta\,\mathbb{1}\{d \in \text{post}\} \cdot \mathbb{1}\{p \in \text{crit}\} + \alpha_u + \phi_p + \delta_d + \varepsilon_{u,d,p},$$

with critical hours as treated and flat hours as the within-day falsifier. Table reports θ.

	March (IDA reform)		October (DA reform)	
DiD θ	DA	IDA	DA	IDA
σ_p (EUR/MWh)	**4.17*****	0.10	**0.68*****	-0.68^{*}
$\hat{\alpha}$ (EUR/MWh)	$-$**40.2*****	11.8^{*}	**28.3*****	6.0^{**}
$\hat{\beta}$ (EUR/MWh per MW)	**0.27*****	-0.88^{*}	$-$**0.15*****	0.20
$\hat{\gamma}$ ($\times 10^3$)	$-$**3.99*****	10.0	**1.03*****	7.6
HHI_p	$-$**0.076*****	$-$**0.041*****	$-$**0.059*****	**0.051*****

- **Main finding (green):** DA at October. σ_p rises, $\hat{\alpha}$ lifts, $\hat{\beta}$ flattens, $\hat{\gamma}$ bends convex, HHI_p falls. Bid stack *deconcentrates*: more MW packed cheap at low quantities (Cournot scale-up) plus a steeper tail above the clearing price.
- **Anticipation (orange):** DA at March — opposite curvature sign $\Rightarrow$ different mechanism (anticipating the IDA reform).
- **Null findings (red):** σ_p and $\hat{\gamma}$ at the March IDA reform; $\hat{\beta}$ at the October IDA. The reformed venue does not restructure on its own reform date.

Residual demand slope — per-firm Cournot competitiveness rises on the reformed venue

Object. For each of the four dominant generators, the slope $b_{m,t}$ of the residual demand it faces (Cournot primitive). Steeper $\Rightarrow$ less unilateral market power.

DiD spec (per firm, per reform $\times$ venue).

$$\log b_{f,t} \;=\; \theta_f \, \mathbb{1}\{t \in \text{post}\} \;+\; \text{month FE} \;+\; (\text{wind, solar daily}) \;+\; \varepsilon_{f,t}.$$

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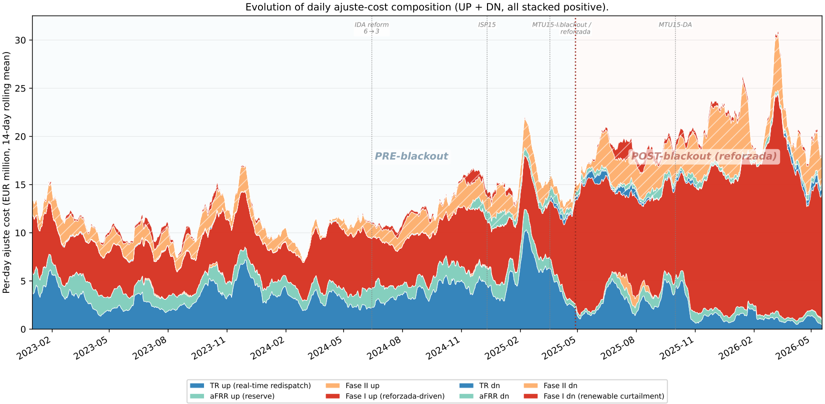
- **Negative finding (red):** DA at October. The newly-reformed DA does NOT thicken its own slope. Effect is venue-specific to the IDA side.

The April 2025 blackout triggered a separate regulatory regime



- Post-blackout, the system operator keeps combined-cycle gas plants synchronised for voltage support, paid through a separate redispatch channel (+5M EUR/day). Daily technical-restrictions cost: 2.4 → 10.7 M EUR.
- **Negative finding:** the real-time TR/aFRR cost decrease is largely *timing displacement* from real-time to Fase I within the post-blackout regime, *not* a granularity efficiency gain.

Part 4

Conclusion

Conclusion

Thank you.

Questions?

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Part 6

Appendix

ID15 price effect at longer post-windows

	40 d†	**90 d**	**180 d**	**196 d**‡
OLS daily, ID15 IDA price (EUR/MWh, Newey-West HAC lag=7)				
base (month + DOW)	-32^{***}	-37^{***}	-25^{***}	-25^{***}
+ year-by-renew (per year)	-46^{***}	-53^{***}	-54^{***}	-53^{***}
+ year-by-renew (2024+25 pooled)	$\mathbf{-39}^{***}$	$\mathbf{-42}^{***}$	$\mathbf{-30}^{***}$	$\mathbf{-30}^{***}$
OLS hourly, ID15 IDA price (EUR/MWh, Newey-West HAC lag=24×7)				
base (hour + month + DOW)	-34^{***}	-38^{***}	-25^{***}	-25^{***}
+ year-by-renew (per year)	-40^{***}	-43^{***}	-31^{***}	-31^{***}
+ year-by-renew (2024+25 pooled)	$\mathbf{-30}^{***}$	$\mathbf{-35}^{***}$	$\mathbf{-24}^{***}$	$\mathbf{-24}^{***}$
BSTS daily, ID15 IDA price (per-year long-pre spec)				
Real effect	-34^{*}	-35	-12	-10

† *Pre-blackout headline window.* ‡ *Up to DA15 cutover (max ID15 post not contaminated by DA15). All specs significant at $p < 0.01$ on the OLS rows. ID15 DA cross-market estimates are within ± 1 EUR/MWh of the ID15 IDA own-venue values at every cell.*

Price effect by hour class

OLS hourly + year-by-renewable (2024+25 pooled) spec, pre-window 2022-01-01 onward. Newey-West HAC lag $= 24 \times 7$. Same headline post-windows (ID15 40 d, DA15 92 d).

Hour class	ID15 IDA	ID15 DA cross	DA15 DA	DA15 IDA cross
Morning ramp (5–8)	-33^{**}	-32^{*}	3	1
Evening ramp (16–22)	-25^{*}	-24^{*}	1	-1
Midday (11–14)	$-\mathbf{39}^{***}$	$-\mathbf{37}^{***}$	-3	-5
Flat (1–3)	-34^{**}	-30^{*}	2	3

Reading. **ID15 price drop is broadly uniform across hour classes** (-25 to -39 EUR/MWh) and slightly larger at midday (when solar saturation pushes prices low, the granular leg helps integrate flexibility most). **DA15 null everywhere** — no hour-class heterogeneity rescues the headline DA15 null.

Placebo prices table

Same row structure as the headline prices table; cutovers shifted to *2024-03-19* (ID15 placebo, 40 d post) and *2024-10-01* (DA15 placebo, 92 d post). Same pre-window 2022-01-01 onward.

	ID15 placebo (cut. 2024-03-19)		**DA15 placebo (cut. 2024-10-01**	
	IDA	DA cross	DA	IDA cross
OLS daily — base (month + DOW)	-49^{***}	-50^{***}	10	12
OLS daily — + year-by-renewable (per year)	2	1	50^{***}	50^{***}
OLS daily — + year-by-renewable (2024+25 pooled)	2	1	50^{***}	50^{***}
OLS hourly — base (hour + month + DOW)	-49^{***}	-49^{***}	9	11
OLS hourly — + year-by-renewable (per year)	-11	-12	47^{***}	48^{***}
OLS hourly — + year-by-renewable (2024+25 pooled)	-11	-12	47^{***}	48^{***}
BSTS daily	-22	-15	4	27

Per-IDA-session price effect breakdown

Pre-window 2024-06-14 onward (European-IDA 3-session regime); year-by-renewable 2024+25 pooled spec; Newey-West HAC SE (lag $= 7$ daily, 24×7 hourly).

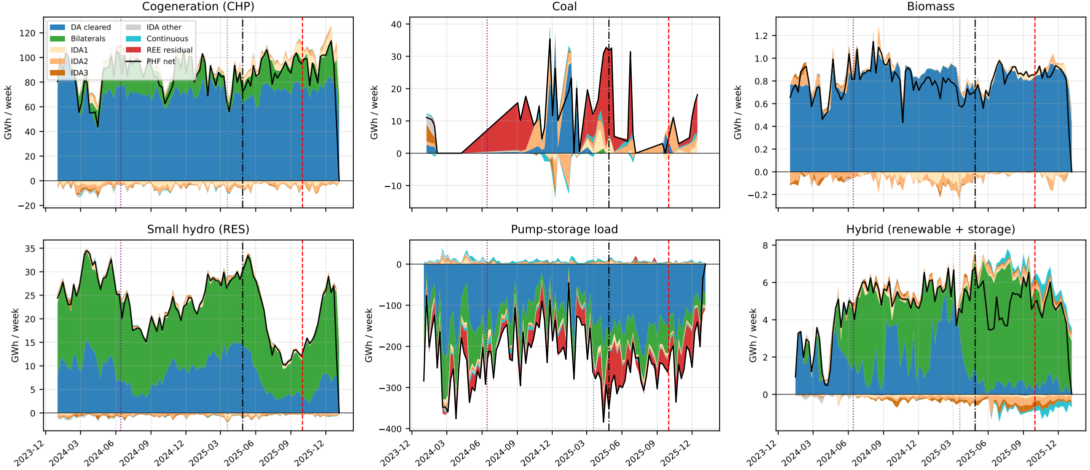
Session	S1 (early)	S2 (mid)	S3 (late)
ID15 own-venue effect (EUR/MWh) — OLS daily	$-\mathbf{37}^{***}$	$-\mathbf{35}^{***}$	$-\mathbf{29}^{***}$
ID15 own-venue effect (EUR/MWh) — OLS hourly	$-\mathbf{39}^{***}$	$-\mathbf{37}^{***}$	$-\mathbf{34}^{***}$
DA15 cross-market effect — OLS daily	14	17^{*}	11
DA15 cross-market effect — OLS hourly	18	23^{*}	13

Reading. ID15 price drop is uniform across sessions (-34 to -39 EUR/MWh under hourly) — the granular-leg discount is not concentrated in any one session. DA15 cross-effect on IDA sessions is positive but mostly null, consistent with the DA-own-venue null on the headline slide.

Price wedges placebos

Program cascade for other technologies

Other technologies --- weekly program cascade (supply above 0, consumption / down-redispatch below 0). Pump-storage load is purely consumption (all below 0). Black line = PHF net.



Appendix — In-band offered energy migrates to the granular market

		ID15 (Net, MWh/class-hr)		DA15 (Net, MWh/class-hr)	
Tech	**Hour class**	DA	IDA	DA	IDA
CCGT	morning ramp	77	**2090*****	**3462****	$-$**1627*****
	midday	-1497^{***}	**1426*****	988^{**}	773^{*}
	evening ramp	191	**1859*****	**2458*****	-152
	flat	-863^{**}	**1405*****	**2966****	$-$**1157*****
Pump-storage	midday	151	**979****	**433****	**868****
	evening ramp	-1	**775***	251	-408

CHECK HERE WHAT WE ARE DOING!!!

- Under ID15, CCGT and pump-storage push energy *into* IDA (the granular leg); under DA15, they push it back into DA.
- Hydro is the exception (reservoir-driven, 2024 placebo poor); not leaned on.
- Bid-side primitive of the cleared scale-up in critical hours under DA15.

Appendix — Bid-shape parallel trends (visual check)